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COURSE CODE: SBT-DF203
LAB NAME: SMTP EMAIL TRAFFIC FORENSICS
DATE: 14/09/2026
INSTRUCTOR: AMINU IDRIS

1. Source Materials and Slide Alignment

- Email Traffic Forensics: SMTP purpose, ports 25/587/465, plaintext commands, STARTTLS/SMTPS distinctions, Base64 encoding, stream reassembly and email-content analysis.
- The slide practical uses smtp.pcap containing Microsoft Outlook traffic and requires timestamps, client, content, ports, IP addresses and MAC addresses.
- The exercise treats credentials as sensitive evidence and requires protected reporting.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical SMTP packet capture has been supplied for analysis. You must determine when the email exchange occurred, identify the client and server, reconstruct SMTP commands and response codes, decode relevant Base64 authentication fields, recover the message content, and document the network path without exposing sensitive data unnecessarily.

3. Learning Outcomes

- Explain the role of SMTP and distinguish plaintext SMTP, STARTTLS and implicit TLS.
- Identify SMTP commands, server response codes and authentication exchanges.
- Reassemble an SMTP TCP stream and reconstruct message headers and body.
- Decode Base64 values in a controlled offline manner.
- Extract timestamps, client software, IP addresses, ports and MAC addresses.
- Assess evidential limitations when encryption is used or capture is incomplete.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Treat any recovered username, password or message content as confidential assessment evidence. Mask sensitive values in screenshots unless the instructor explicitly requires the full value.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

6. Lab Folder Structure and Evidence Preparation

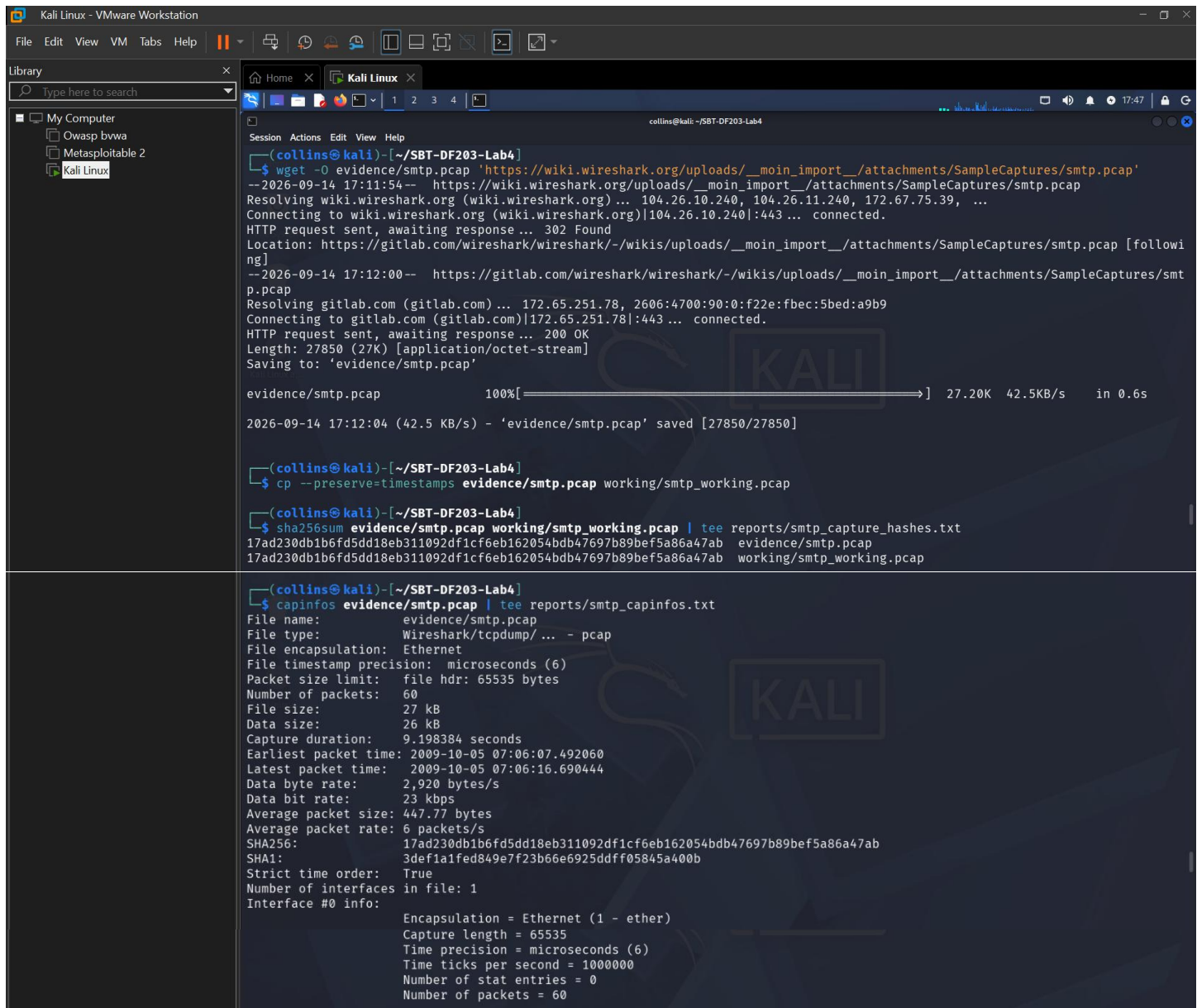
```
mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y wireshark tshark python3
wget -O evidence/sntp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/sntp.pcap'
cp --preserve=timestamps evidence/sntp.pcap working/sntp_working.pcap
sha256sum evidence/sntp.pcap working/sntp_working.pcap | tee reports/sntp_capture_hashes.txt
capinfos evidence/sntp.pcap | tee reports/sntp_capinfos.txt
```

Screenshot required: sntp.pcap file details, capinfos summary and matching original/working-copy hashes.

ANSWER:



```
(collins@kali)~/SBT-DF203-Lab4
$ wget -O evidence/sntp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/sntp.pcap'
--2026-09-14 17:11:54-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/sntp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 104.26.10.240, 104.26.11.240, 172.67.75.39, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.10.240|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/sntp.pcap [following]
--2026-09-14 17:12:00-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/sntp.pcap
Resolving gitlab.com (gitlab.com)... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/sntp.pcap'

evidence/sntp.pcap 100%[=====] 27.20K 42.5KB/s in 0.6s

2026-09-14 17:12:04 (42.5 KB/s) - 'evidence/sntp.pcap' saved [27850/27850]

(collins@kali)~/SBT-DF203-Lab4
$ cp --preserve=timestamps evidence/sntp.pcap working/sntp_working.pcap

(collins@kali)~/SBT-DF203-Lab4
$ sha256sum evidence/sntp.pcap working/sntp_working.pcap | tee reports/sntp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/sntp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/sntp_working.pcap

(collins@kali)~/SBT-DF203-Lab4
$ capinfos evidence/sntp.pcap | tee reports/sntp_capinfos.txt
File name: evidence/sntp.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 07:06:07.492060
Latest packet time: 2009-10-05 07:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925ddf05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = Ethernet (1 - ether)
Capture length = 65535
Time precision = microseconds (6)
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 60
```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab4-COLLINS AMADI
Trainee name	COLLINS AMADI
Date and time started	14/09/2026 04:48:21
Evidence file name(s)	evidence/smtp.pcap
Source or generation method	KALI LINUX COMMAND
Original SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
Analysis workstation/VM	COLLINS@KALI
Notes on any changes	No target file changes. cp --preserve=timestamps used for integrity preservation.

8. Part A - Inventory the Capture and Locate SMTP Streams

```
tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
```

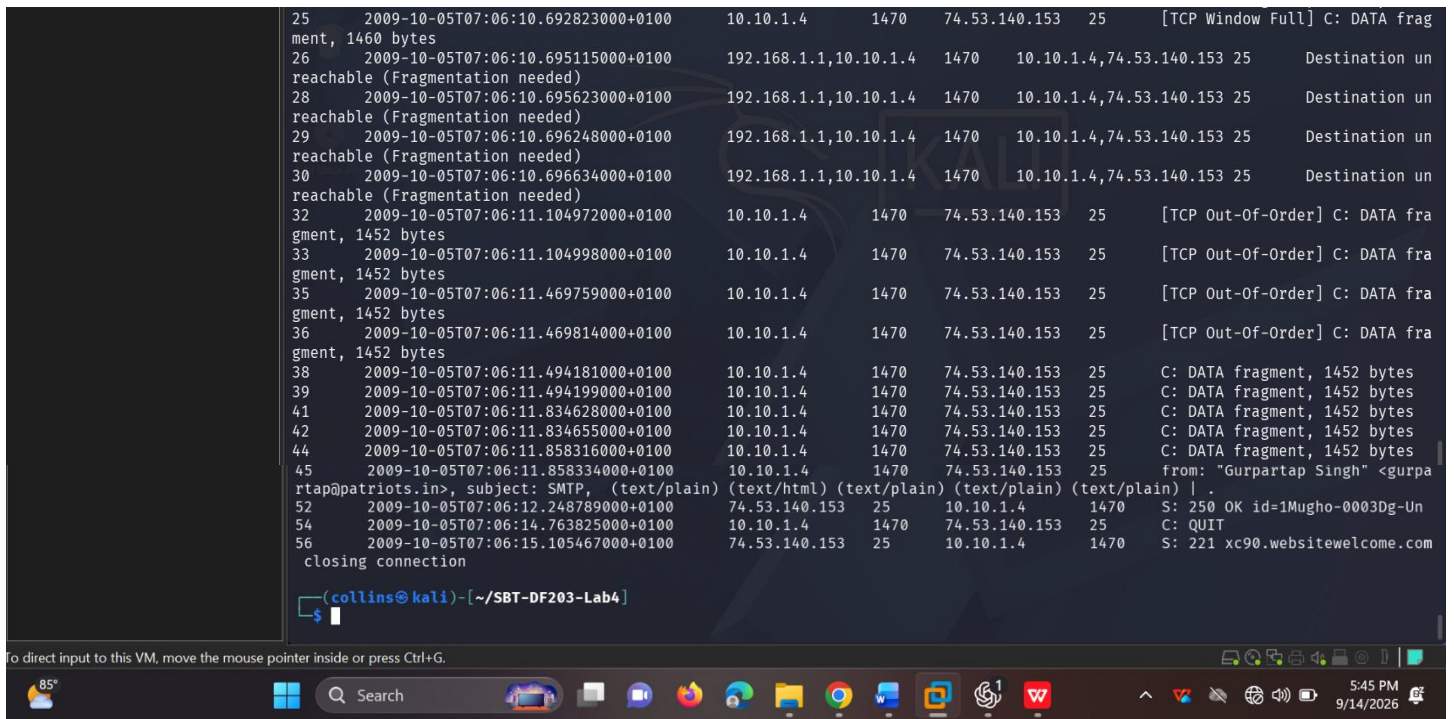
```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
| tee reports/smtp_packet_inventory.tsv
```

Record the SMTP server port, client ephemeral port, first packet time, last packet time and number of SMTP frames.

ANSWER:

Duration	Frames	Bytes	Start
10.10.1.4:1470 ↔ 74.53.140.153:25 000 7.5777	25	1,980 bytes	0.036986

Frame Number	Time	IP Src	TCP Src Port	IP Dst	TCP Dst Port	Info
6	2009-10-05T07:06:08.219663000+0100	10.10.1.4	1470	74.53.140.153	25	S: 220-xc90.websitewelcome.com
7	2009-10-05T07:06:08.224809000+0100	10.10.1.4	1470	74.53.140.153	25	C: EHLO GP
9	2009-10-05T07:06:08.566183000+0100	10.10.1.4	1470	74.53.140.153	25	S: 250-xc90.websitewelcome.com
10	2009-10-05T07:06:08.568729000+0100	10.10.1.4	1470	74.53.140.153	25	C: AUTH LOGIN
11	2009-10-05T07:06:08.911081000+0100	10.10.1.4	1470	74.53.140.153	25	S: 334 VXNlcmShbWU6
12	2009-10-05T07:06:08.911655000+0100	10.10.1.4	1470	74.53.140.153	25	C: User: Z3VycGFydGFWQHBhdHJpb
13	2009-10-05T07:06:09.253544000+0100	10.10.1.4	1470	74.53.140.153	25	S: 334 UGFzc3dvcmQ6
14	2009-10-05T07:06:09.254118000+0100	10.10.1.4	1470	74.53.140.153	25	C: Pass: cHVuamFiQDEyMw==
15	2009-10-05T07:06:09.613798000+0100	10.10.1.4	1470	74.53.140.153	25	S: 235 Authentication succede
16	2009-10-05T07:06:09.614414000+0100	10.10.1.4	1470	74.53.140.153	25	C: MAIL FROM: <gurpartap@patri
17	2009-10-05T07:06:09.956765000+0100	10.10.1.4	1470	74.53.140.153	25	S: 250 OK
18	2009-10-05T07:06:09.957250000+0100	10.10.1.4	1470	74.53.140.153	25	C: RCPT TO: <raj_deol2002in@ya
19	2009-10-05T07:06:10.319708000+0100	10.10.1.4	1470	74.53.140.153	25	S: 250 Accepted
20	2009-10-05T07:06:10.320203000+0100	10.10.1.4	1470	74.53.140.153	25	C: DATA
21	2009-10-05T07:06:10.661679000+0100	10.10.1.4	1470	74.53.140.153	25	S: 354 Enter message, ending w
22	2009-10-05T07:06:10.692743000+0100	10.10.1.4	1470	74.53.140.153	25	C: DATA fragment, 1460 bytes
23	2009-10-05T07:06:10.692786000+0100	10.10.1.4	1470	74.53.140.153	25	C: DATA fragment, 1460 bytes
24	2009-10-05T07:06:10.692804000+0100	10.10.1.4	1470	74.53.140.153	25	C: DATA fragment, 1460 bytes



9. Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to client	220	Service ready	7t2009-10-05T07:06:08.224809000+0100t10.10.1.4t74.53.140.153tEHLOtGPtt
2	Client to server	EHLO/HELO	Client greeting	9t2009-10-05T07:06:08.566183000+0100t74.53.140.153t10.10.1.4ttt250txc90.websi\ntewelcome.com Hello GP
3	Client to server	AUTH	Authentication method	10t2009-10-05T07:06:08.568729000+0100t10.10.1.4t74.53.140.153tAUTHtLOGINtt
4	Client to server	MAIL FROM	Envelope sender	16t2009-10-05T07:06:09.614414000+0100t10.10.1.4t74.53.140.153tMAILtFROM:<gurpar\ntap@pa\trio\ts.in>tt
5	Client to server	RCPT TO	Envelope recipient	18t2009-10-05T07:06:09.957250000+0100t10.10.1.4t74.53.140.153tRCPTtTO:<raj_deol2002in@yahoo.co.in>tt
6	Client to server	DATA	Message content begins	20t2009-10-05T07:06:10.320203000+0100t10.10.1.4t74.53.140.153tDATAttt
7	Client to server	QUIT	Session termination	54t2009-10-05T07:06:14.763825000+0100t10.10.1.4t74.53.140.153tQUITttt

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.response.parameter \
-E header=y -E separator=t \
| tee reports/smtp_commands_responses.tsv
```

10. Part C - Decode Base64 Authentication Evidence

Base64 is encoding, not encryption. Copy only the required encoded values into the offline script. Do not use online decoders.

```
python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
```

PYLAB

Reporting Rule

In the public-facing report body, show a masked value such as analyst@example.com and P*****3. Place full decoded values only in the protected evidence appendix if required by the instructor.

ANSWER:

```
(collins@kali) ~/SBT-DF203-Lab4
$ python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycG9ydGFWQHBhdHJpb3RzLmlu',
    'password': 'cHVuamF1QDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
```

PYLAB
username: gurpartap@patriots.in
password: punjab@123

11. Part D - Reconstruct the Email Message

1. In Wireshark select an SMTP packet and choose Follow > TCP Stream.
2. Identify SMTP envelope fields separately from RFC message headers.
3. Record Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer where present.
4. Determine whether the body is plain text, HTML, multipart or contains an attachment.
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt.

```
# Replace STREAM with the relevant stream number
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM \
| tee reports/smtp_stream_STREAM.txt
```

```
grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
```

12. Part E - Determine Client, Hosts and Network Metadata

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv
```

```
tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
```

ANSWER:

Session	Actions	Edit	View	Help
(collins@kali) ~/SBT-DF203-Lab4				
\$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \				
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \				
-e ip.dst -e tcp.dstport -e tcp.stream \				
tee reports/smtp_network_metadata.tsv				
6	1254722768.219663000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
7	1254722768.220090000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
8	1254722768.220517000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
9	1254722768.220944000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
10	1254722768.221371000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
11	1254722768.221798000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
12	1254722768.222225000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
13	1254722768.222652000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
14	1254722768.223079000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
15	1254722768.223506000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
16	1254722768.223933000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
17	1254722768.224360000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
18	1254722768.224787000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
19	1254722768.225214000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
20	1254722768.225641000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
21	1254722768.226068000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
22	1254722768.226495000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
23	1254722768.226922000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
24	1254722768.227349000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
25	1254722768.227776000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
26	1254722768.228203000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
27	1254722768.228630000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
28	1254722768.229057000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0
29	1254722768.229484000	00:1f:33:d9:81:60	00:e0:1c:3c:17:c2	74.53.140.153 25 10.10.1.4 1470 0

```
29 1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
30 1254722770.696634000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
31 1254722771.104972000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
32 1254722771.104998000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
33 1254722771.469759000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
34 1254722771.469914000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
35 1254722771.494181000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
36 1254722771.494199000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
37 1254722771.834628000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
38 1254722771.834655000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
39 1254722771.858316000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
40 1254722771.858334000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
41 1254722772.248789000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
42 1254722774.763825000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
43 1254722775.105467000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0

(collins@kali) ~/SBT-DF203-Lab4
$ tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
tee Reports/client_indicators.tsv
22 0 Line-based text data (47 lines)
26 0 Line-based text data (12 lines)

(collins@kali) ~/SBT-DF203-Lab4
$
```

13. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested and whether subsequent payload becomes TLS.
- If traffic is encrypted, document metadata that remains visible: endpoints, ports, timing, certificate/session information and byte counts.
- Do not claim message content or credentials when the capture does not reveal them.
- Explain that port number alone does not prove encryption; protocol negotiation and packet dissection must be examined.

ANSWER:

The server advertises 250-STARTTLS in response to the initial client handshake. However, the client completely ignores this upgrade capability and falls straight into explicit unencrypted text authentication using AUTH LOGIN.

Because the transport channel remains strictly text-based, 100% of payload content, data envelopes, file layers, and passwords were intercepted cleanly without requiring key injection or decryption scripts.

Port assignment configurations (like port 25, 465, or 587) act solely as functional traffic assumptions. A forensic investigator cannot declare a stream encrypted based entirely on port numbers. Protocol state negotiation sequences, deep packet inspection (DPI), and structure verification inside tools like tshark must prove cryptographic handshakes took place.

14. Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	2009-10-05 07:06:07.492060 2009-10-05 07:06:16.690444
Client IP/MAC and port	
Server IP/MAC and port	
SMTP server banner	250-xc90.websitewelcome.com
Client software	
Authentication method	AUTH LOGIN
Envelope sender and recipient	MAIL FROM: <gurpartap@patriots.in>
Message From/To/Subject	RCPT TO: <raj_deol2002in@yahoo.co.in>
Message body type	354 Enter message, ending with "." on a line by itself
Attachment present?	Yes (Plaintext attachment asset named test.txt)
STARTTLS/TLS observed?	No. Advertised by the server host but rejected/bypassed by the client.
Key limitations	Capture isolates localized sender transaction parameters only; it provides no evidence of final mailbox storage or server-to-recipient transit status.